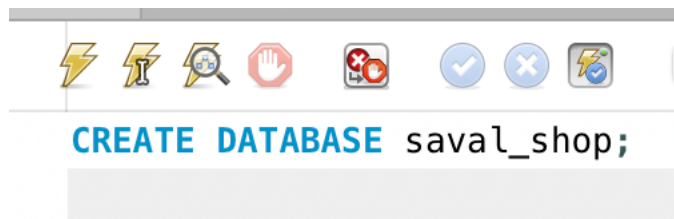


Practice Set 1: Create and Insert (Q1–Q5)

1. Create the shop database and the products table exactly as shown.

Confirm with SHOW TABLES;.

CODE SCREENSHOT:



DATABASE SCREENSHOT:

Result Grid		Filter Rows: Search
	Database	
☐	DataAnalytics	
☐	fashionconnect	
☐	fashionconnectDB	
☐	information_schema	
☐	lavanya_fashion	
☐	medical_record_system	
☐	mysql	
☐	Parks_and_Recreation	
☐	performance_schema	
☐	saval_shop	

SQL PRACTICE

2. Insert all 6 products in one single INSERT statement.

CODE SCREENSHOT:

```
CREATE TABLE products (  
  product_id INT PRIMARY KEY ,  
  pname VARCHAR (50) ,  
  price INT ,  
  category VARCHAR (30) ,  
  stock INT  
);  
  
INSERT INTO products  
VALUES  
(1, 'Rice Bag', 2200, 'Grocery', 50),  
(2, 'Laptop', 85000, 'Electronics', 10),  
(3, 'Mobile', 30000, 'Electronics', 25),  
(4, 'Noodles', 25, 'Grocery', 500),  
(5, 'Shampoo', 350, 'Beauty', 80),  
(6, 'TV', 55000, 'Electronics', 8);
```

DATABASE SCREENSHOT:

21 • SELECT*FROM products;
22

100% 36:19

Result Grid Filter Rows: Search

	product_...	pname	price	category	stock	
	1	Rice Bag	2200	Grocery	50	
	2	Laptop	85000	Electronics	10	
	3	Mobile	30000	Electronics	25	
	4	Noodles	25	Grocery	500	
	5	Shampoo	350	Beauty	80	
	6	TV	55000	Electronics	8	
	NULL	NULL	NULL	NULL	NULL	

SQL PRACTICE

3. Insert product 7, giving only product id and pname ('Pen'). Which columns become NULL?

CODE SCREENSHOT:

```
INSERT INTO products (product_id, pname)
VALUES (7, 'Pen');
```

DATABASE SCREENSHOT:

25 • `SELECT * FROM products;`

100% 22:25

Result Grid Filter Rows: Search

	product_id	pname	price	category	stock
1	1	Rice Bag	2200	Grocery	50
2	2	Laptop	85000	Electronics	10
3	3	Mobile	30000	Electronics	25
4	4	Noodles	25	Grocery	500
5	5	Shampoo	350	Beauty	80
6	6	TV	55000	Electronics	8
7	7	Pen	NULL	NULL	NULL

4. Try to insert another product with product id = 3. Write down the exact error and explain why it happened.

CODE SCREENSHOT:

27 • `INSERT INTO products (id, pname)`
28 `VALUES (3, 'Keyboard');`

100% 1:27

Action Output

	Time	Action	Response
11	20:42:05	SELECT FROM products LIMIT 0, 10	7 row(s) returned
12	20:44:26	INSERT INTO products (id, pname) VALUES (3, 'Keyboard')	Error Code: 1054. Unknown column 'id' in 'field list'

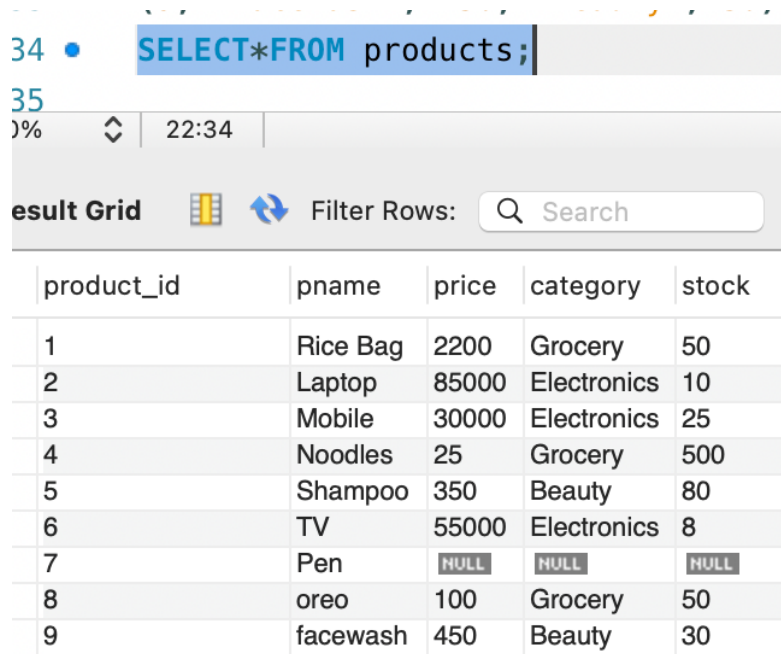
Error Explanation: *This error happened because primary key is product id here and it must be unique, there can't be duplicate product id because product id 3 is already there.*

5. Insert two more products of your choice in one statement: one Grocery item and one Beauty item.

CODE SCREENSHOT

```
INSERT INTO products
VALUES
(8, 'oreo', 100, 'Grocery', 50),
(9, 'facewash', 450, 'Beauty', 30);
```

DATABASE SCREENSHOT:



The screenshot shows a database interface with a SQL query editor at the top and a results grid below. The query is `SELECT * FROM products;`. The results grid displays the following data:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
2	Laptop	85000	Electronics	10
3	Mobile	30000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
6	TV	55000	Electronics	8
7	Pen	NULL	NULL	NULL
8	oreo	100	Grocery	50
9	facewash	450	Beauty	30

Practice Set 1: Read (Q6–Q15)

6. Show all columns of all products.

CODE SCREENSHOT:

```
SELECT * FROM products;
```

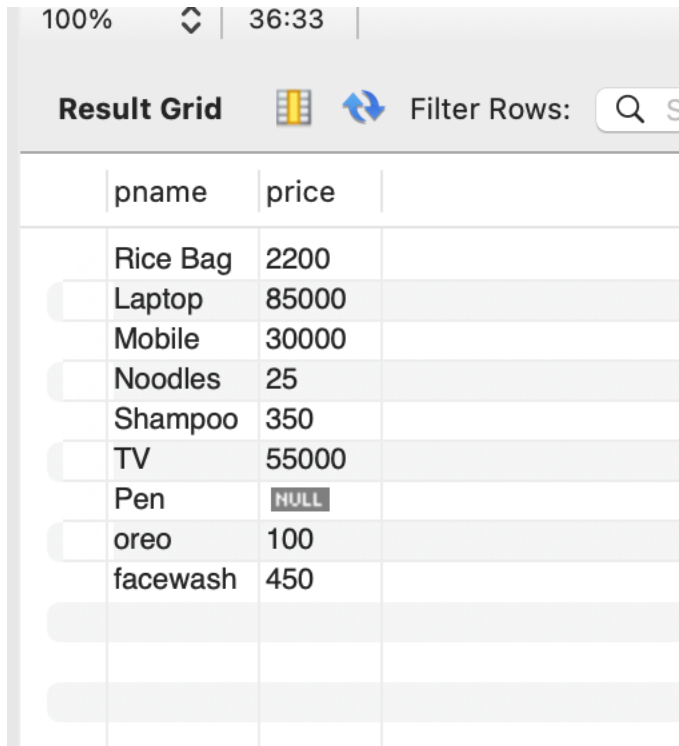
DATABASE SCREENSHOT:

Result Grid   Filter Rows: Search					
	product_id	pname	price	category	stock
	1	Rice Bag	2200	Grocery	50
	2	Laptop	85000	Electronics	10
	3	Mobile	30000	Electronics	25
	4	Noodles	25	Grocery	500
	5	Shampoo	350	Beauty	80
	6	TV	55000	Electronics	8
	7	Pen	NULL	NULL	NULL
	8	oreo	100	Grocery	50
	9	facewash	450	Beauty	30
	NULL	NULL	NULL	NULL	NULL

7. Show only pname and price.

CODE SCREENSHOT:

```
SELECT pname, price FROM products;
```

DATABASE SCREENSHOT:

100% | 36:33

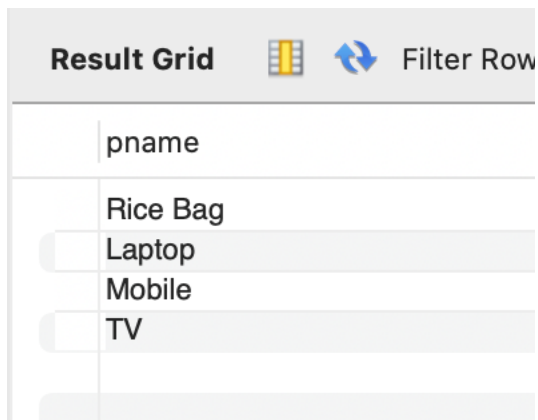
Result Grid | Filter Rows:

	pname	price	
	Rice Bag	2200	
	Laptop	85000	
	Mobile	30000	
	Noodles	25	
	Shampoo	350	
	TV	55000	
	Pen	NULL	
	oreo	100	
	facewash	450	

8. Show products with price above 1000.

CODE SCREENSHOT:

```
SELECT pname FROM products WHERE price>1000;
```

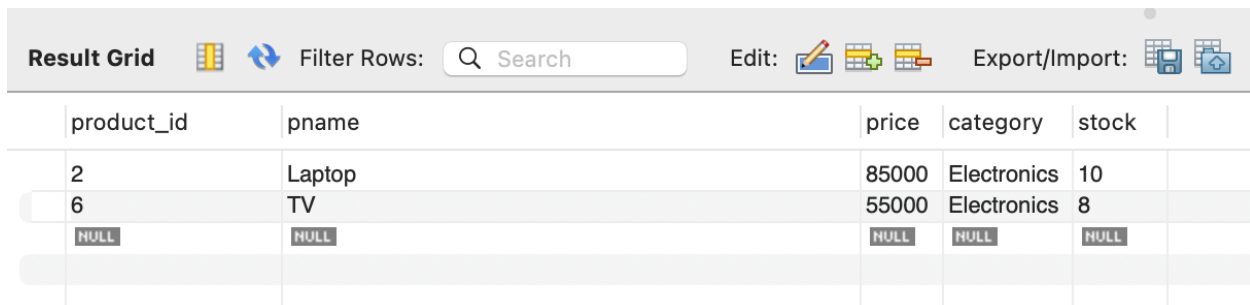
DATABASE SCREENSHOT:

product_id	pname	price	category	stock
1	Rice Bag	1000	Food	100
2	Laptop	85000	Electronics	10
3	Mobile	15000	Electronics	5
4	TV	55000	Electronics	8

9. Show Electronics products with stock less than 20 (AND).

CODE SCREENSHOT:

```
SELECT * FROM products WHERE category = 'Electronics' AND stock < 20;
```

DATABASE SCREENSHOT:

product_id	pname	price	category	stock
2	Laptop	85000	Electronics	10
6	TV	55000	Electronics	8
NULL	NULL	NULL	NULL	NULL

SQL PRACTICE

10. Show products that are Grocery or cost less than 500 (OR).

CODE SCREENSHOT:

```
SELECT*FROM products WHERE category = 'Grocery' OR price < 500;
```

DATABASE SCREENSHOT:

Result Grid   Filter Rows: Search Edit:    Export/Import: 						
	product_id	pname	price	category	stock	
	1	Rice Bag	2200	Grocery	50	
	4	Noodles	25	Grocery	500	
	5	Shampoo	350	Beauty	80	
	8	oreo	100	Grocery	50	
	9	facewash	450	Beauty	30	
	NULL	NULL	NULL	NULL	NULL	

11. Show products in categories Grocery and Beauty, using IN.

CODE SCREENSHOT:

- ```
SELECT*FROM products WHERE category IN ("Grocery", "Beauty");
```

### DATABASE SCREENSHOT:

| Result Grid   Filter Rows: <input type="text" value="Search"/> Edit:    Export/Import:  |            |          |       |          |       |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|-------|----------|-------|--|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | product_id | pname    | price | category | stock |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1          | Rice Bag | 2200  | Grocery  | 50    |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4          | Noodles  | 25    | Grocery  | 500   |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5          | Shampoo  | 350   | Beauty   | 80    |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 8          | oreo     | 100   | Grocery  | 50    |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 9          | facewash | 450   | Beauty   | 30    |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | NULL       | NULL     | NULL  | NULL     | NULL  |  |

12. Show products with price between 300 and 40000. Is 350 included? Why?

**CODE SCREENSHOT:**

```
SELECT * FROM products WHERE price BETWEEN 300 AND 40000;
```

**DATABASE SCREENSHOT:**

| Result Grid   Filter Rows: <input type="text" value="Search"/> Edit:    Export/Import:  |          |       |             |       |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|-------------|-------|--|
| product_id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | pname    | price | category    | stock |  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rice Bag | 2200  | Grocery     | 50    |  |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mobile   | 30000 | Electronics | 25    |  |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Shampoo  | 350   | Beauty      | 80    |  |
| 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | facewash | 450   | Beauty      | 30    |  |
| NULL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | NULL     | NULL  | NULL        | NULL  |  |

13. Show products whose name starts with M.

**CODE SCREENSHOT:**

```
SELECT * FROM products WHERE pname LIKE 'M%';
```

**DATABASE SCREENSHOT:**

| Result Grid   Filter Rows: <input type="text" value="Search"/> Edit:    Export/Import:  |        |       |             |       |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|-------------|-------|--|
| product_id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | pname  | price | category    | stock |  |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Mobile | 30000 | Electronics | 25    |  |
| NULL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | NULL   | NULL  | NULL        | NULL  |  |

## SQL PRACTICE

14. Show products whose name contains the letter o.

### CODE SCREENSHOT:

```
SELECT * FROM products WHERE pname LIKE "%o%";
```

### DATABASE SCREENSHOT:

| Result Grid | Filter Rows: | Search | Edit:       | Export/Import: |
|-------------|--------------|--------|-------------|----------------|
| product_id  | pname        | price  | category    | stock          |
| 2           | Laptop       | 85000  | Electronics | 10             |
| 3           | Mobile       | 30000  | Electronics | 25             |
| 4           | Noodles      | 25     | Grocery     | 500            |
| 5           | Shampoo      | 350    | Beauty      | 80             |
| 8           | oreo         | 100    | Grocery     | 50             |
| NULL        | NULL         | NULL   | NULL        | NULL           |

15. Show the 2 most expensive products (which two keywords together?).

### CODE SCREENSHOT:

```
SELECT * FROM products ORDER BY price DESC LIMIT 2;
```

### DATABASE SCREENSHOT:

| Result Grid | Filter Rows: | Search | Edit:       | Export/Import: |
|-------------|--------------|--------|-------------|----------------|
| product_id  | pname        | price  | category    | stock          |
| 2           | Laptop       | 85000  | Electronics | 10             |
| 6           | TV           | 55000  | Electronics | 8              |
| NULL        | NULL         | NULL   | NULL        | NULL           |

## Practice Set 1: Update and Delete (Q16–Q20)

16. The Laptop price dropped to 79000. Update it using its id.

### CODE SCREENSHOT:

```
UPDATE products SET price = 79000 WHERE product_id = 2;
```

### DATABASE SCREENSHOT:

| Result Grid   Filter Rows: <input type="text" value="Search"/> Edit:    Export/Import:  |          |       |             |       |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|-------------|-------|--|
| product_id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | pname    | price | category    | stock |  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rice Bag | 2200  | Grocery     | 50    |  |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Laptop   | 79000 | Electronics | 10    |  |

17. Festival offer: reduce the price of every Electronics product by 10%.

Hint: SET price = price \* 0.9.

### CODE SCREENSHOT:

```
UPDATE products SET price = price * 0.9 WHERE product_id IN (2, 3, 6);
```

### DATABASE SCREENSHOT:

60 • `SELECT pname, price FROM products WHERE c`

| Result Grid   Filter Rows: <input type="text" value="Search"/> Export: |       |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--|--|
| pname                                                                                                                                                                                                                                        | price |  |  |
| Laptop                                                                                                                                                                                                                                       | 71100 |  |  |
| Mobile                                                                                                                                                                                                                                       | 27000 |  |  |
| TV                                                                                                                                                                                                                                           | 49500 |  |  |

18. Ten Rice Bags were sold. Decrease its stock by 10 with one query.

**CODE SCREENSHOT:**

```
UPDATE products SET stock = stock - 10 WHERE pname = "Rice Bag";
```

**DATABASE SCREENSHOT:**

| Result Grid |       | Filter Rows: | Search | Export: |
|-------------|-------|--------------|--------|---------|
| pname       | stock |              |        |         |
| Rice Bag    | 40    |              |        |         |

19. Delete the product named Pen. Then delete all products with stock below 10. How many rows did each query remove?

**CODE SCREENSHOT:**

```
DELETE FROM products WHERE pname = 'Pen';
```

- Pen is removed which was id no 7 so 1 row affected here

```
DELETE FROM products WHERE stock < 10;
```

- Here also 1 row affected because one product only has stock less than 10 which is TV.

## SQL PRACTICE

### DATABASE SCREENSHOT:

| product_id | pname    | price | category    | stock |
|------------|----------|-------|-------------|-------|
| 1          | Rice Bag | 2200  | Grocery     | 40    |
| 2          | Laptop   | 71100 | Electronics | 10    |
| 3          | Mobile   | 27000 | Electronics | 25    |
| 4          | Noodles  | 25    | Grocery     | 500   |
| 5          | Shampoo  | 350   | Beauty      | 80    |
| 8          | oreo     | 100   | Grocery     | 50    |
| 9          | facewash | 450   | Beauty      | 30    |
| NULL       | NULL     | NULL  | NULL        | NULL  |

20. Run SELECT to check the final table. Then answer: which command would empty the table but keep it, and which would remove the table completely?

### CODE SCREENSHOT:

```
70 • SELECT * FROM products;
```

00% 24:70

Result Grid Filter Rows: Search Edit: Export/Import:

| product_id | pname    | price | category    | stock |
|------------|----------|-------|-------------|-------|
| 1          | Rice Bag | 2200  | Grocery     | 40    |
| 2          | Laptop   | 71100 | Electronics | 10    |
| 3          | Mobile   | 27000 | Electronics | 25    |
| 4          | Noodles  | 25    | Grocery     | 500   |
| 5          | Shampoo  | 350   | Beauty      | 80    |
| 8          | oreo     | 100   | Grocery     | 50    |
| 9          | facewash | 450   | Beauty      | 30    |
| NULL       | NULL     | NULL  | NULL        | NULL  |

## COMMANDS AND THEIR WORK:

- **DELETE:** *this key word will remove only rows but not the table like: it will empty the product table but doesn't delete the product table.*
- **DROP:** *this keyword will remove all the row including the table which means it will remove all the row and the product table too from the database*

*THANK\_YOU!!*